

Electrical Engineering Department

EE104L: Computer Aided Engineering Project

Automatic Fan

Controller with temperature sensor.

Submitted by:

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Context:

- INTRODUCTION
- WORKING
- CIRCUIT ANALYSIS
- PCB LAYOUT
- ENCLOSURE DESIGN
- FEATURES
- COMPONENTS
- APPLICATIONS
- CONCLUSION
- RFERENCE

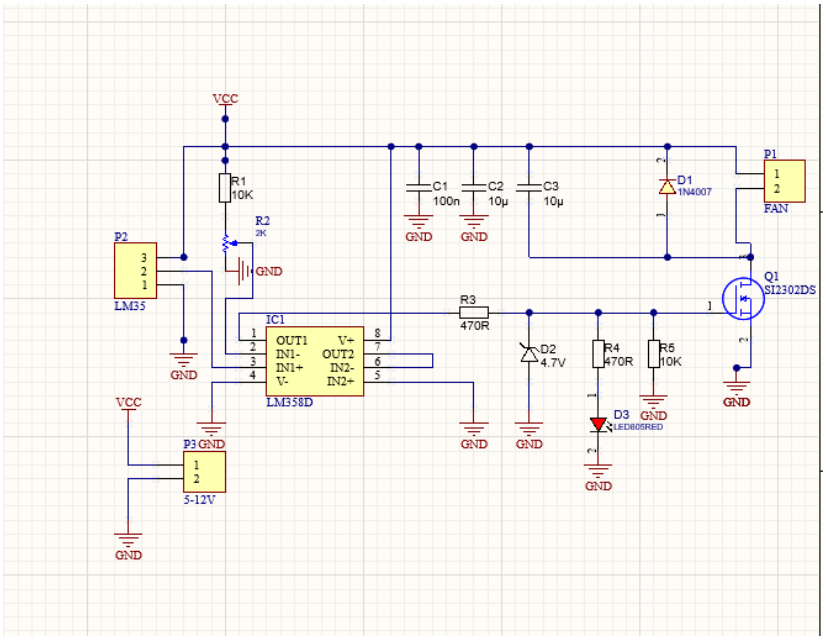
Introduction:

Smart homes are the future of home automation. They offer greater control and convenience to homeowners by allowing them to remotely control various aspects of their homes, such as lighting, heating, security, and entertainment systems, through their smartphones or other devices. One of the most popular aspects of a smart home is its ability to control the lighting system. With smart lighting, homeowners can create customized lighting scenarios, set schedules, and even adjust the brightness and color temperature of the lights. In this project, we will design a smart lighting system that can be controlled through a smartphone application using Bluetooth technology. The system will consist of a microcontroller, Bluetooth module, and a relay board that will interface with the lighting system. With this system, homeowners will be able to control their lighting system from anywhere in the house, eliminating the need for physical switches and providing a more convenient and efficient way of controlling the lights.

Working:

The LM358 op-amp (U1) is configured as a comparator with one input connected to a thermistor, which is used to sense the temperature of the environment. The other input is connected to a potentiometer (R2), which is used to set the reference temperature at which the output of the comparator switches. When the temperature sensed by the thermistor is higher than the reference temperature set by R2, the output of U1 goes high, and when it is lower, the output goes low. The output of U1 is connected to the gate of a MOSFET (Q1), which acts as a switch to control the power to the load. The MOSFET is chosen for its low on-resistance and high current handling capability. The LED (D3) is used as an indicator to show the status of the output. R3 and D2 are used to limit the voltage at the gate of Q1 and protect it from overvoltage. R4 is fixed to 470 ohms to maintain the voltage across D3 at around 4.7 volts. C1 and C2 are bypass capacitors that reduce noise in the circuit, and C3 reduces switching noise.

Schematic Diagram:



Design rule check:

Altium Designer

Design Rule Verification Report

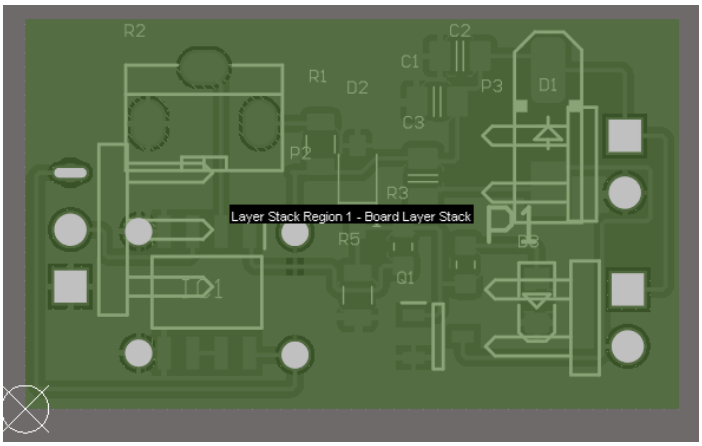
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Time:	8:27:56 PM	**Rule Violations:**	0
Elapsed Time:	00:00:02		
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Summary

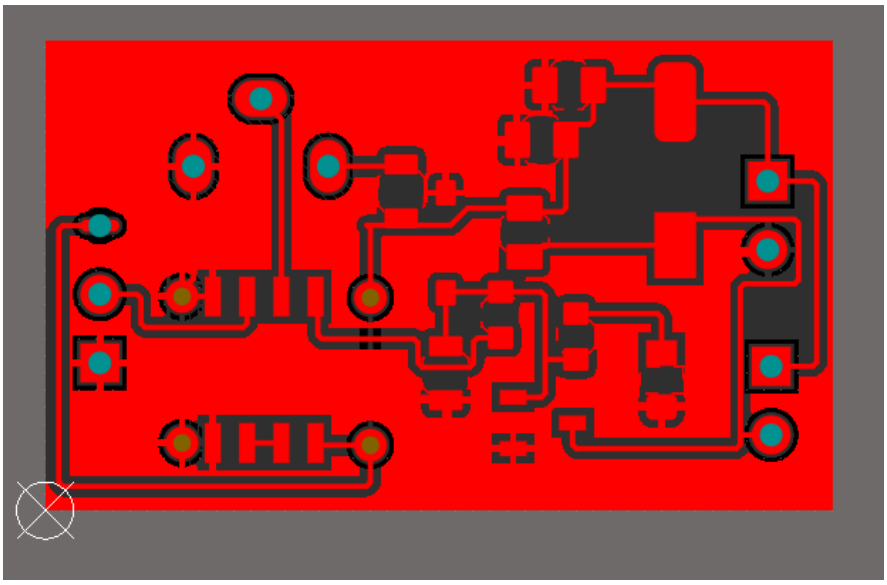
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PCB Layout:

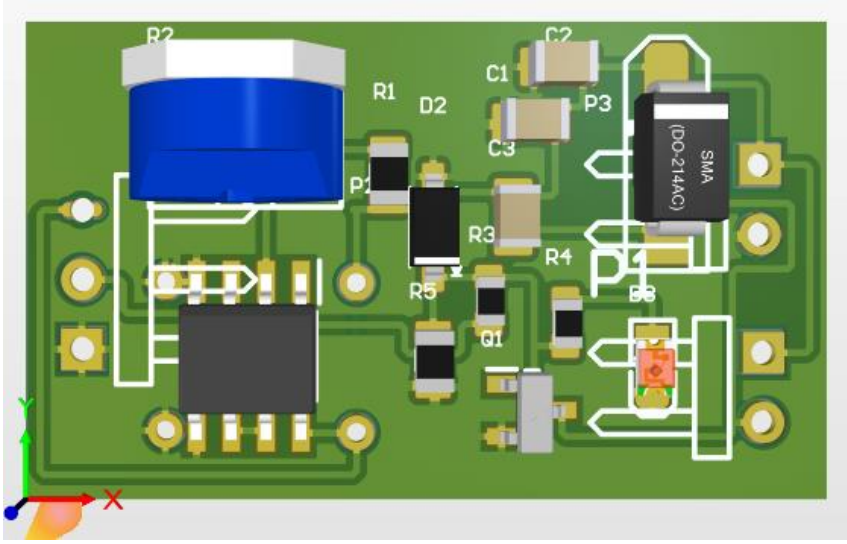
Board shape:



Routing layout



3rd Design:

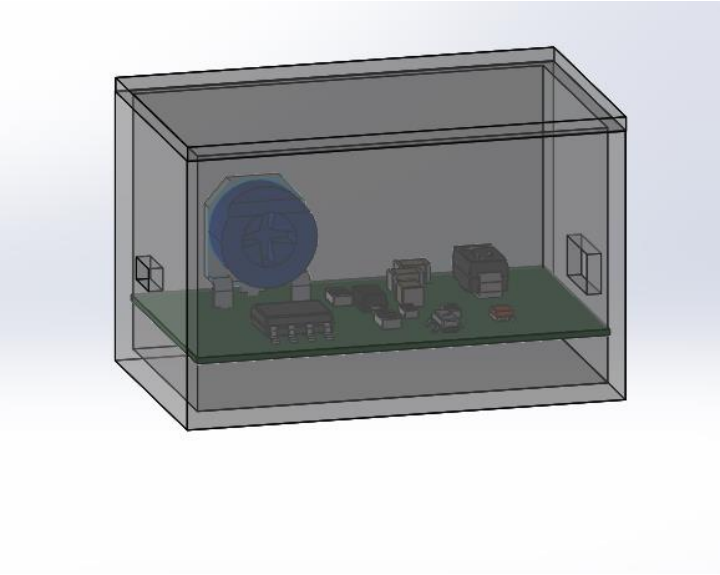


SOILD WORKS

Enclosure Design:

Working:

Firstly, you need to identify how to use a sketch. In my design, I used a front view sketch. Then, I created a rectangle shape and added dimensions of 30mm for length and 19mm for width. After that, I went to the extruded box and set the device's measurements. I inserted this shape into my step file, which I had created for the printed circuit board (PCB). Next, I created an upper surface on this shape and reduced its width slightly. Finally, I placed this last shape on top of the rectangle box.



Features:

- Supply 12V DC (Range 5V to 12V DC)
- Fan Load Up to 1Amp Continues (Maximum Load 4Amps)
- Temperature Threshold Range Adjustable 10 to 100 Degree Centigrade using Multiturn Trimmer Potentiometer.
- Simple and Low-Cost Temperature Management Solution for Heatsink or Equipment
- On-Board Trimmer Potentiometer to Adjust the Temperature Threshold

Components required:

Values	Designator	Quantity
• 100nF-50V	C1	1
• 10uF-16V	C2,C3	2
• 1N4007	D1	1
• 4.7V-Zener SOD-80	D2	1

• LED-R	D3	1
• XH-2P 2.54mm Pitch	P1,P3	2
• XH-3P P2 2.54mm Pitch	P2	1
• Si2302DS SOT-23	Q1	1
• 10K-0805	R1,R5	2
• 2K 3296W Pot	R2	1
• 470R-0805	R3,R4	2
• LM358-SOIC8	U1	1

Application:

- ❖ The circuit serves as a temperature-controlled switch
- ❖ It can be applied in heating and cooling systems, refrigeration systems, and temperature-sensitive equipment
- ❖ It provides automatic temperature control to maintain the desired temperature range
- ❖ It can prevent damage to the equipment
- ❖ It can be integrated into home automation systems to control temperature and turn on/off heating or cooling systems based on temperature readings
- ❖ The circuit has a wide range of applications in various industries
- ❖ It can be customized to meet specific requirements.

Conclusion:

In conclusion, the Cooling FAN Controller using an LM35 project provides a simple and effective solution to the problem of high temperatures of power components in commercial and home appliances. The automatic FAN controller board can be easily embedded inside devices, and the user can easily set the temperature threshold using a

potentiometer. The proposed board is small and compact, making it ideal for use in a variety of applications.

References:

[1] LM358 datasheet: https://www.st.com/resource/en/datasheet/lm358.pdf

[2] https://www.javatpoint.com/tree-vs-graph-data-structure

[3]https://www.electronics-lab.com/project/cooling-fan-controller-using-temperature-sensor-lm35/

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